

Market Regimes & Adaptive Portfolio Construction

Author – Arun Ramesh

ID – 2025A7PS0633P

OBSERVATIONS:

VIX vs Time plot:

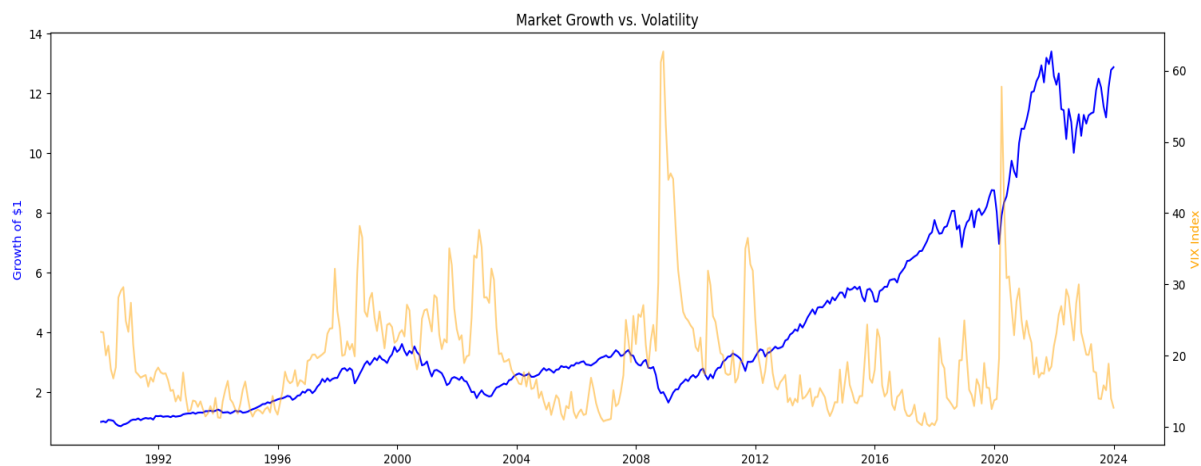
VIX predicts the volatility of the market over the next 30 days using option prices.

During periods of stress, we can see that the VIX spikes which means that the market expects to be volatile over the next 30 days. For example, 2008 (global financial crisis), 2020 (COVID-19).



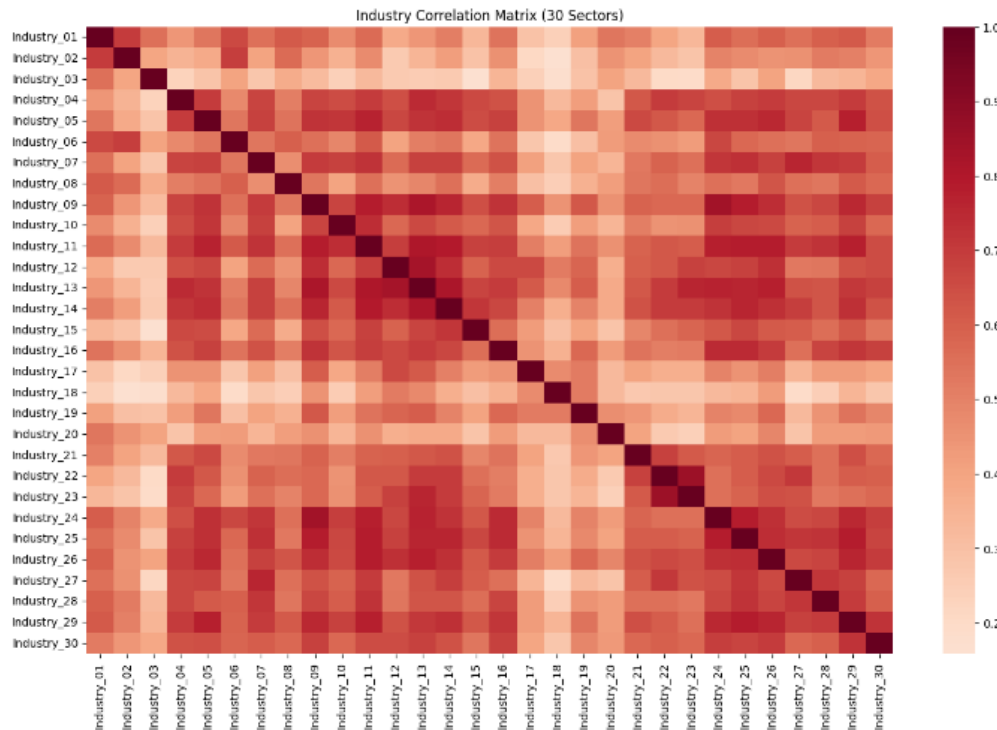
Market Growth and VIX:

We can observe that when VIX spikes, market growth generally falls down. This indicates periods of stress where people panic sell all their assets and put their money into “safe havens” like gold or bonds.



Correlation Heat-Map:

This shows how much various industries move together. A higher correlation means that the industries move in the same direction while a lower means that they move in opposite directions. The range of correlation varies from -1 to 1.

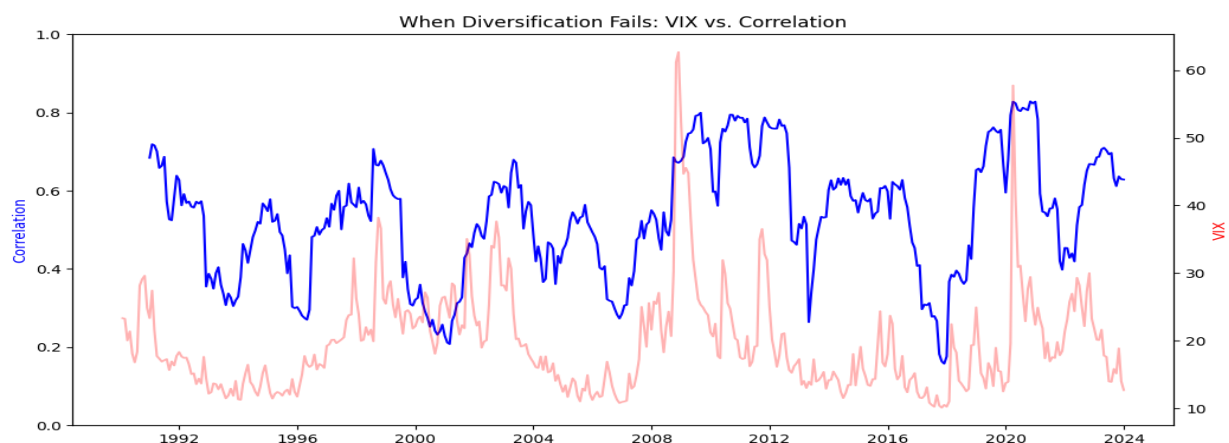


VIX vs Correlation:

During periods when VIX rises, correlations also tend to rise. This usually means that the market is in stress and all industries are moving down together.

This is where diversification fails as all markets are going down and correlations are high, so here you would ideally want to invest in safe assets. Diversification would work best when the correlations are near 0, where there is no relation between industries. So some industries do well, while the others perform neutral.

Here an average rolling correlation of 12 month window is used in the plot.



HYPOTHESIS:

The market acts differently in various regimes, hence we should be able to identify the various latent regimes in the market and then design a suitable strategy for each regime.

For example, when the market is stressed, we should have only a percentage of our capital invested in the market as the market is highly volatile and unpredictable so as to reduce the risk we are exposing ourselves to. During other periods, we should diversify our portfolio to avoid the downfall of a certain industry hurting the growth of the money we've invested.

METHODOLOGY:

1.0 TERCILE REGIME: (Percentile-Based)

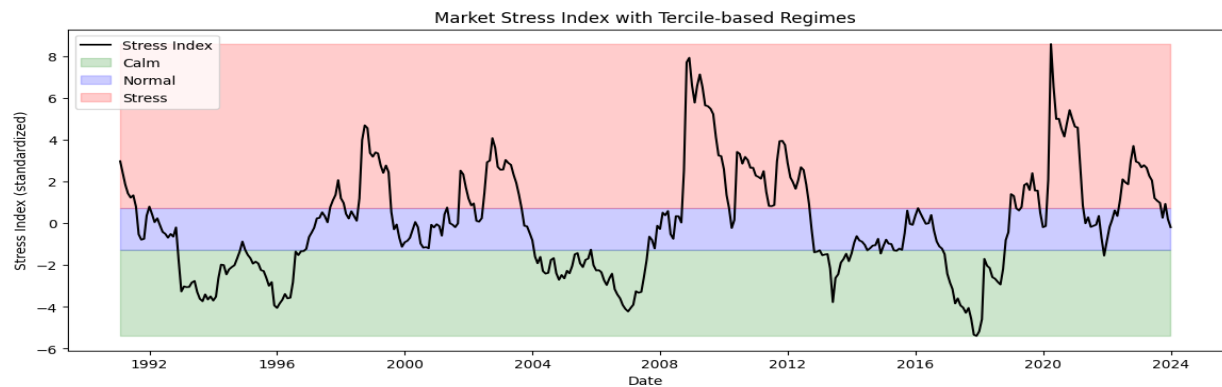
- 1.1** The first regime detection approach was to use VIX, volatility, and rolling average cross-industry correlation to define a stressed regime. If the value of all these features were in the top 30th percentile, then it would be considered a stressed regime, otherwise calm regime.

The problem with this model was that the mean values of VIX, volatility and correlation were still relatively high for the calm regime and I thought that there could be a regime with lower values of these features. So it seemed logical to add another regime in between which is called the Normal Regime.

- 1.2** The second regime detection approach was to make 3 equal periods of each regime (calm, normal, stress) using only VIX. The problem with this approach was that I took only 1 parameter which decided the regime and this resulted in other features such as volatility, correlation to have similar values across different regimes.

As a result, a stress index was created which took VIX, volatility and correlation as its features and then was equally divided into 3 periods of calm, normal, and stress regimes. This showed good identifiable economical differences across the 3 regimes.

Stress index plot along with shaded regimes based on tercile regime detector:



I thought that having only one regime detector could result in many false decisions and be inaccurate, so I decided to make another regime detector so both models could compliment each other and help to create an overall more efficient regime detector.

Hidden Markov Model (HMM):

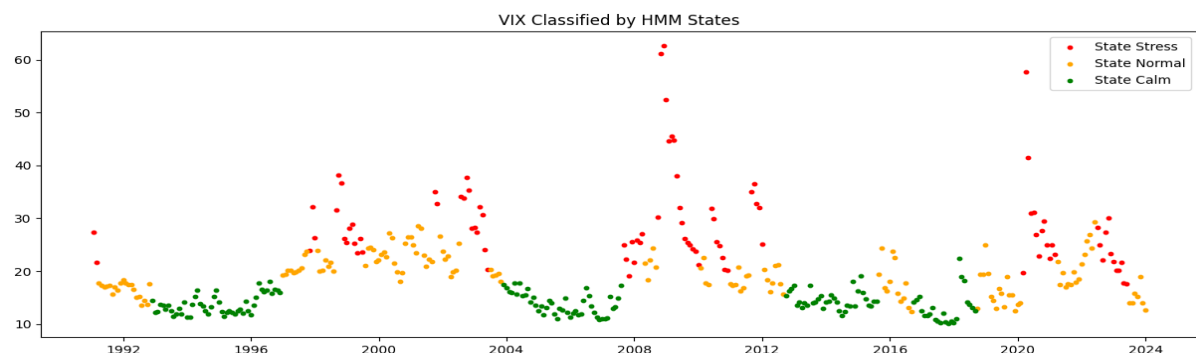
The second regime detection model was implemented using a Hidden Markov Model. The features of the model are:

- Market volatility
- Rolling average correlation
- Stress index

Various number of hidden states were tested. A three-state model was selected as it provided distinct regimes with clear economic and meaningful differences in volatility and correlation behaviour. Models with more number of states produced regimes which seemed identical while ones with lesser number of states wasn't showing the entire picture. I named the 3 states as calm, normal and stress according to the values of volatility, VIX, correlation etc.

The HMM outputs a probability of each month being in a certain regime, the highest probability regime was selected.

VIX Classified by HMM States plot:



FINAL COMBINED REGIME DETECTOR:

I thought that tercile regime reacts quickly to market changes (possibly giving false signals), while HMM performs smooth transitions between regimes (possibly identifying a regime slower or completely missing it).

I had tried shifting the time period of the tercile regime a few months ahead to check if both the regimes had a higher percentage of agreements which would prove that tercile regime reacts a little earlier than the HMM Model, but this wasn't true.

The percentage agreements of the regimes between both the models are 78%.

To create the final combined regime:

- If both models agree, that regime is selected.
- If they disagree, the riskier regime is selected.

[Risk: stress > normal > calm]

I noticed that when the models disagreed on stress regime, the features of this disagreement was very similar to when they both agreed on normal regime. Hence I included this disagreement into the normal regime as well.

PORTFOLIO CONSTRUCTION:

It seemed logical to control your exposure based on different regimes, as having high exposure during stressed periods could expose you to severe drawdowns. Specific exposure ranges for each regime have been decided, the exact value of exposure depends on the stress index which consists of VIX, volatility and correlation.

1. Discrete Industry Buckets:

Industries were grouped into 3 buckets namely, (defensive, neutral, cyclic) and weights were distributed equally within each bucket based on a cyclicity score whose features are:

- Market beta
- Average returns during stress periods
- Volatility

Market beta is the sensitivity of industry returns relative to the overall market returns. It tells us about the direction and amplitude of movements w.r.t. the market.

The weights of the buckets were chosen such that during calm periods, cyclic group would have the highest weight. While during stressed periods, defensive group would get the highest weight.

The drawback of this method is that:

- The assumption of equal group sizes can be incorrect
- Hard cutoff can separate 2 industries with similar behaviour
- Weights of industry in each bucket is equal.

I didn't think that this method of distributing weights is effective as there is a very hard classification between groups of industries. The idea of the best and worst industry in a particular bucket getting equal weight didn't seem logical to me. I wanted a method where the best industry for a particular regime should get the most weight while the worst should get the least weight, and so that the distribution of weight is continuous.

2. Continuous Regime-Conditional Ranking:

To avoid hard classification between industries, a continuous weighting system was designed

For each regime, a score was assigned to each industry based on the features that are required and essential for an industry to have in that regime.

- Calm: Main focus on returns, with small volatility penalty.
- Normal: Balanced weights of returns, volatility and VIX beta.
- Stress: Mainly trying to protect capital using volatility, max drawdown, CvaR, VIX beta, and volatility transmission.

Max Drawdown: This is the maximum observed loss that your portfolio suffers.

CVaR 5%: This tells you the average loss your portfolio suffers in the worst 5% of months.

VIX Beta: This measures the sensitivity of industry returns to the changes in the level of VIX of the market.

Volatility transmission: I wanted to obtain the VIX for individual industries, with this I could easily identify how each industry reacts when overall market VIX changes. Since VIX is a 30-day forward looking indicator, by using the market VIX, we can check how volatile the industry is relative to the market VIX by looking at the next months returns. This could be used to identify industry groups too!

- Low Volatility Transmission: Likely to be defensive industries.
- High Volatility Transmission: Likely to be cyclic industries.

3. Benchmarks:

Two benchmarks were used:

- Equal-weight industry portfolio
- Market Portfolio

Both these markets maintain full exposure and do not adapt to regimes.

RESULTS:

This section evaluates how portfolios behave under different market regimes, rather than focusing on returns.

REGIME CONDITIONAL VOLATILITY:

Across the regimes, risk is managed differently

- **Calm Regime:**

Both discrete and continuous regime-based portfolios have higher volatility when compared with static benchmarks. This is the result of having greater exposure during calm periods towards industries which include more risk.

- **Normal Regime:**

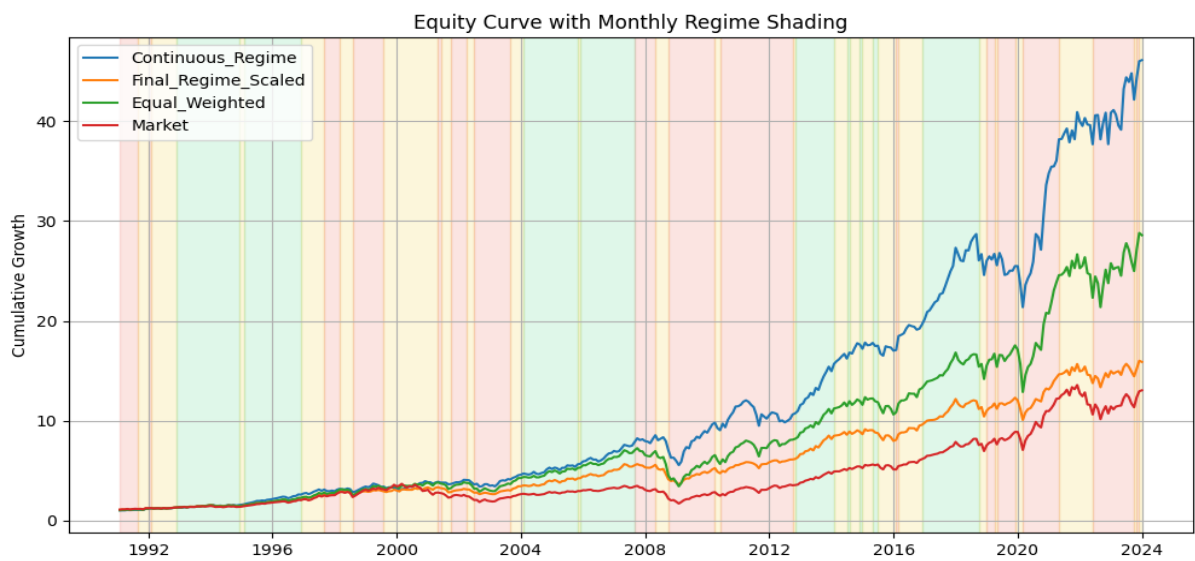
Volatility is lowest for both discrete and continuous regime-based portfolios when compared with static benchmarks. This is due to the reduced exposure and more constraints on volatility during selection of industries during weighting process.

- **Stress Regime:**

Volatility is the lowest for both discrete and continuous regime-based portfolios again. Which is an indicator of effective risk management during stressed periods.

Regime-aware weight allocation does not minimise volatility everywhere, but it prevents volatility from exploding when it matters most.

Equity Curve with Monthly Regime Shading:



Statistics of Final Portfolios:

	Discrete Portfolio	Continuous Portfolio	Equal Weighted	Market
Annual Return (%)	9.15	13.11	11.48	8.98
CAGR (%)	8.75	13.05	10.69	8.09
Annual Volatility (%)	12.04	12.58	15.90	15.27
Sharpe	0.76	1.04	0.72	0.59
Max Drawdown (%)	-36.73	-36.08	-52.96	-54.16

RISK ANALYSIS:

This section displays the “worst-case” performances of the strategy. It is a result of market panic when correlations rise and volatility spikes.

	Discrete Portfolio	Equal Weighted	Market	Continuous Portfolio
5% VaR (%)	-5.14	-6.38	-7.60	-4.82
1% VaR (%)	-9.62	-11.12	-10.36	-9.26
Worst Month (%)	-13.34	-20.43	-17.20	-11.47
Max Drawdown (%)	-36.73	-52.96	-54.16	-36.08

The ability to improve Max Drawdown and VaR is due to the regime-awareness, without knowing which market regime we likely are in, we wouldn't have been able to control this parameter. During stressed periods we are essentially reducing our exposure and shifting our money into defensive industries.

The portfolios created have significantly lesser drawdown, VaR values as when compared to the benchmark portfolios. This shows that identifying the latent regime and creating a suitable strategy for that regime benefits us.

FAILURES AND LEARNINGS:

The portfolio underperforms during regime transitions into stress.

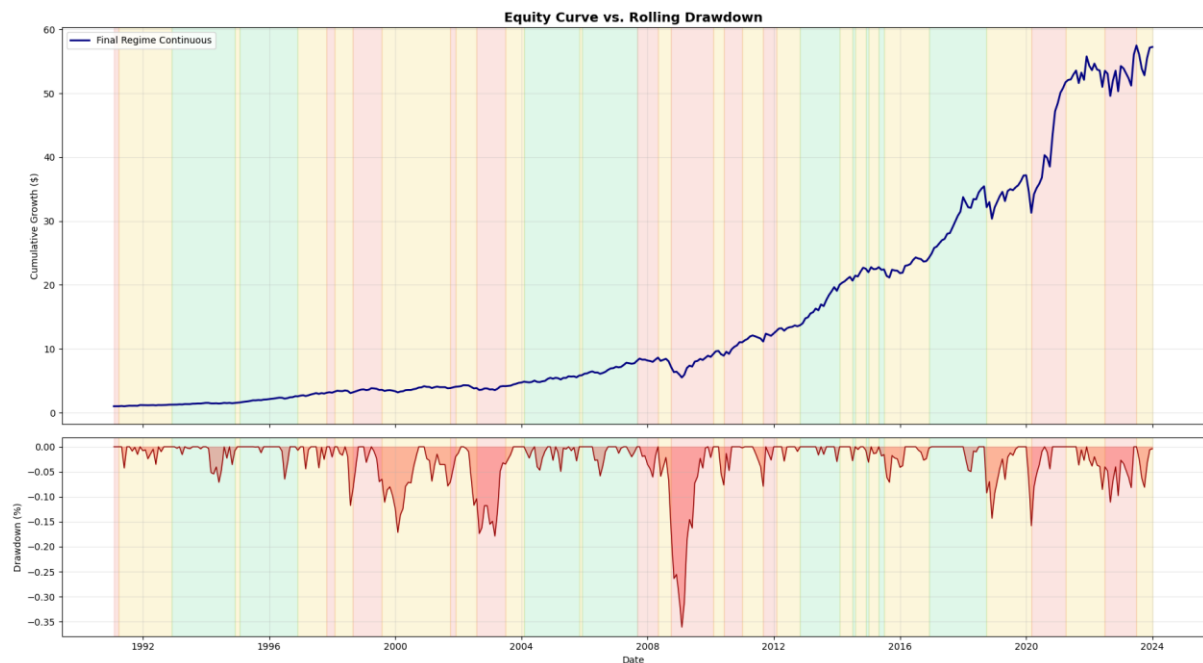
- **Models are slow to react:**

The Regime-Detection models primarily rely on VIX, rolling correlation and rolling volatility, which often spikes only after the price has already collapsed. Rolling indicators also respond with a delay which adds to the point. Essentially the price is collapsing while the model still shows that the regime is calm. In this period the model has very high exposure in the market and since correlations spike to 1, the portfolio suffers a massive loss.

The exposure or weights could be adjusted considering the rate of change of parameters instead of just face value for features like VIX, volatility or correlations. This could potentially help increase the speed of detection but could also produce some false signals.

We can see in the plot that the portfolio suffers massive losses during regime transitions

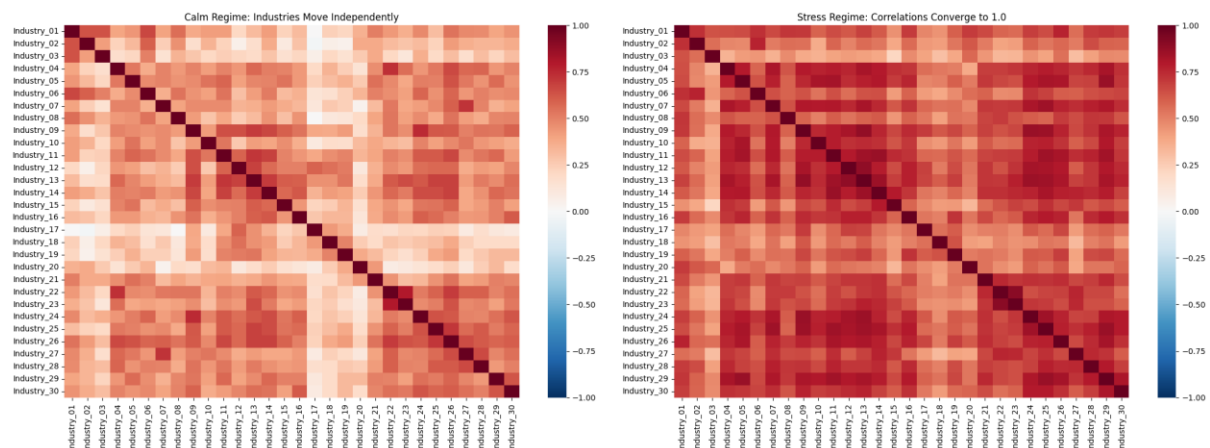
Plot for Continuous Portfolio:



- **Diversification in Stress Regimes:**

In general diversification in stressed markets work fine, when you focus on investing in industries which have low volatility, low market beta during that period (defensive industries) and reducing your exposure, **BUT** during extreme events like 2008 crisis or COVID-19. All correlation spike to 1 and no type of diversification will work.

Here the ideal choice would not be how to diversify but rather to decide how much money is going to be invested into the market (exposure) and to focus more on putting your money into so called “safe havens”



During stressed periods we can see that the correlations are much higher than during calm periods.

Average Correlation (Calm): 0.41

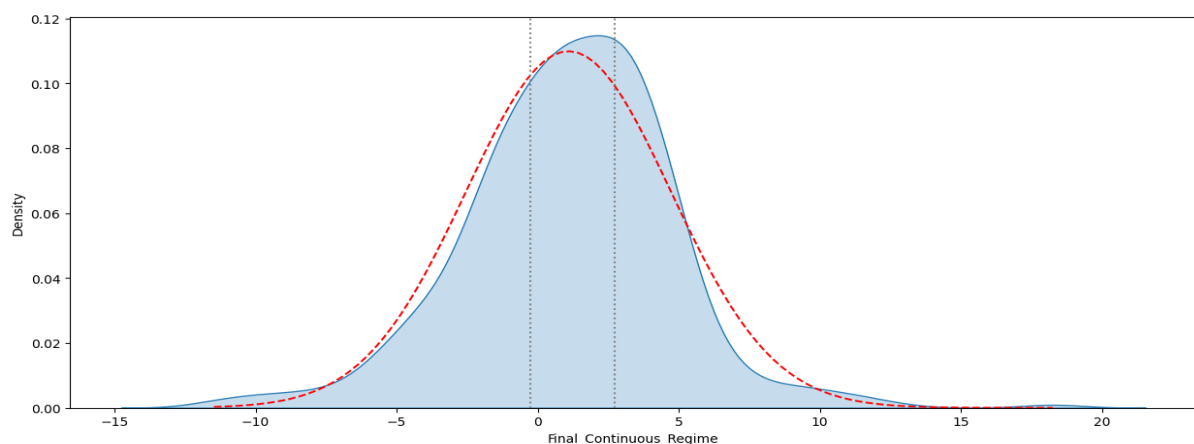
Average Correlation (Stress): 0.64

- **Equal Time Regime Percentile Model:**

Percentile based regime assumes that calm, normal and stress regimes occur with equal frequency. In reality “Stress” regime is a tail event (it happens rarely but with extreme intensity). This means that the model overprepares for stress and is missing out on steady growth during calm periods.

This problem could be encountered with combining absolute thresholds to have more economic meaning. Rate of change of features (spikes) could also be included.

The Hidden Markov Model (HMM) could also be used as it distinguishes states only based on the value of the features given. This doesn't assume that there are equal periods of calm, normal and stressed regimes.



Returns (%) are on the horizontal axis, Probability Density on the vertical axis.

The 2 dotted vertical gray lines are the separations between the regimes.

The red dotted line represents the Theoretical Normal Distribution (Gaussian Distribution)

The blue shaded area represents the probability curve of my returns.

- **Recovery Phase**

The HMM model is also slow to return to the calm/normal regime after going through a stressed period. The model created picks the regime with highest probability, this means that even if the probability of stress is 90% or 50%, the stressed regime is still picked but the intensity of stress has changed which is not being noticed here. This means that we are investing in a calm market but reducing the exposure and investing only in safe industries. We essentially are “missing out” on the steady growth which occurs in calm regime.

Instead of making hard classification on the regimes. A continuous regime classification should be created for smoother transitions. Market regimes are better viewed as probability distributions rather than fixed states, just like we noticed in weight distributions. We could also consider the rate of change of parameters instead of looking at the face value.

END